# Jucheng Shen

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## Education

**Rice University** Expected May 2027

*B.A. in Computer Science, Mathematics, and Mathematical Economic Analysis, Major GPA – 3.7/4.0*

- Relevant Coursework: Reinforcement Learning, Computer Vision, Honors Linear Algebra, Honors ODE, Real Analysis, Differential Geometry, Abstract Algebra, Advanced Algorithms and Data Structures

## Research Experience

**California Institute of Technology** May 2026 – Aug 2026

*Research Intern · Advisor: Prof. Anima Anandkumar*

- Investigated how the spectral structure of continuous language representations affects learning efficiency and generalization in diffusion language models.
- Led the project, developed spectral regularization and attenuation methods, and trained and evaluated models on language modeling, length generalization, translation, and summarization.

**Rice University** Jan 2026 – Sep 2026

*Research Intern · Advisor: Prof. Anastasios Kyrillidis*

- Studied how asymmetric input injection enables a shared-weight recurrent Transformer to develop distinct computational roles for reasoning.
- Proposed *Asymmetric Input Recurrence* (*AIR*) and designed ablation and representation analyses, showing that a clear state-identity signal can induce functional specialization and improve performance on selected abstract reasoning tasks.

**Princeton University** Nov 2025 – Feb 2026

*Research Intern · Advisor: Prof. Zhuang Liu*

- Developed *i1*, a fully open text-to-image model trained from scratch, with systematic exploration of data, architecture, and training choices.
- Built the dataset selection and filtering pipeline and ran 300+ ablation experiments to evaluate modeling and training decisions.

**UT Austin & Intel Labs** May 2025 – Sep 2025

*Research Intern · Advisors: Prof. Atlas Wang & Dr. Souvik Kundu*

- Developed training-free adaptive decoding methods that improve diffusion language model throughput while maintaining generation quality.
- Proposed *OSDT* and led the development of *CadLLM*, designing adaptive decoding strategies and conducting experiments that demonstrated throughput gains of 24–50% and up to 2.3×, respectively.

**Rice University** Apr 2025 – Aug 2025

*Research Intern · Advisor: Prof. Yuke Wang*

- Developed *HyperGen*, a system for generating 2–4K videos through low-resolution generation, upscaling, and tiled refinement without fitting the full-resolution video on a single GPU.
- Implemented *HyperGen* on two baselines, conducted end-to-end evaluations and ablation studies, and contributed figures and manuscript text.

## Publications

[1] **One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer**
**Jucheng Shen***, Barbara Su*, Anastasios Kyrillidis
*NeurIPS 2026*

[2] **Improving the Throughput of Diffusion-based Large Language Models via a Training-Free Confidence-Aware Calibration**
**Jucheng Shen**, Gaurav Sarkar, Yeonju Ro, Sharath Nittur Sridhar, Zhangyang Wang, Aditya Akella, Souvik Kundu
*Findings of ACL 2026*

[3] **Beyond Static Cutoffs: One-Shot Dynamic Thresholding for Diffusion Language Models**
**Jucheng Shen**, Yeonju Ro
*NeurIPS 2025 Efficient Reasoning Workshop*

[4] **i1: A Simple and Fully Open Recipe for Strong Text-to-Image Models**
Boya Zeng, Tianze Luo, Shu Pu, **Jucheng Shen**, Taiming Lu, Gabriel Sarch, Zhuang Liu
*NeurIPS 2026 BeNTo Workshop*

[5] **HyperGen: Empowering Ultra-high-resolution Video Generation Framework with Sketch-Tile Collaboration**
Fanjiang Ye, Zepeng Zhao, Yi Mu, **Jucheng Shen**, Renjie Li, Kaijian Wang, Saurabh Agarwal, Myungjin Lee, Triston Cao, Aditya Akella, Arvind Krishnamurthy, T. S. Eugene Ng, Zhengzhong Tu, Yuke Wang
*PPoPP 2027*

## Teaching Experience

**Rice University** Jan 2026 – Present
*Teaching Assistant · COMP 282: Computational Optimization of AI*

## Academic Service

**Reviewer**

- NeurIPS 2025 Efficient Reasoning Workshop
- COLM 2026 Efficient Reasoning Workshop

## Leadership Experience

**HackRice** Apr 2025 – Oct 2025
*Board Member* (*Tracks and Workshops*)

- Co-designed and shipped the official starter code used by 500+ hackers, and designed competition tracks and judging criteria for HackRice 15.
- Hosted 2 technical workshops for 100+ attendees; ran Q&A and live coding demos.

**Rice Chinese Theater Club** Nov 2023 – Nov 2024
*President*

- Led a team of 30+ members in organizing semester productions, coordinating venues, licensing, sets and props, publicity, and budgeting.
- Performed in three different plays with a combined audience of over 1,000 across all performances, and represented the club at quarterly meetings of the North American College Chinese Theater Alliance.

## Honors & Awards

Caltech Summer Undergraduate Research Fellowship (SURF), California Institute of Technology, 2026.
Frank Liu Jr. Prize for Creative Innovations in Music, Fashion, and the Arts in Napier Rice Launch Challenge 2023.